



PRS Pricing Model

Physical Risk Swap Fair Spread and Basis Calculation

MKM Research Labs

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1 Executive Summary

The Physical Risk Swap (PRS) is a credit-derivative-like instrument that transfers physical flood risk from a protection buyer (e.g. a mortgage lender or property owner) to a protection seller (e.g. a reinsurer or capital markets investor). The PRS pays out upon the occurrence of a defined flood trigger event at a reference gauge or property location, in exchange for a periodic premium (the “spread”).

This document describes the analytical pricing model implemented in the MKM PhysicalRisk platform. The model computes fair PRS spreads using a CDS-equivalent quarterly pricing framework, where the annual flood hazard rate derived from the upstream GEV hazard curve model is converted to a continuous-time Poisson intensity. Fair spreads are calculated as the ratio of the protection leg present value to the risky annuity, expressed in basis points.

A five-component additive basis waterfall adjusts gauge-level PRS prices to property-level prices, accounting for:

1. Model uncertainty (fixed),
2. Distance between the property and its reference gauge(s),
3. Elevation differential,
4. Terrain classification (EA flood zone), and
5. Property composition (type and construction vintage).

The model supports two pricing paths: a lightweight analytical implementation with no external dependencies, and an alternative QuantLib-based implementation using the ISDA CDS engine for cross-validation. Both paths produce consistent results to within rounding tolerance for flat hazard rate inputs.

Key risk: The model assumes a constant (flat) hazard rate over the contract tenor. Time-varying hazard rates arising from climate change trends are not captured. The basis waterfall is additive with no interaction terms between components.

2 Model Purpose and Scope

2.1 Purpose

The PRS pricing model serves three primary functions within the PhysicalRisk platform:

- **Gauge-level PRS pricing:** Compute the fair annual spread (in basis points) for a PRS contract referencing a specific river gauge, given the gauge’s annual flood hazard rate, a chosen trigger level, and contract parameters (tenor, recovery rate, risk-free rate).
- **Property-level PRS pricing:** Adjust gauge-level PRS spreads to reflect the idiosyncratic risk characteristics of individual properties via the five-component basis waterfall.
- **Basis analysis:** Decompose the difference between gauge-level and property-level PRS prices into interpretable components for risk management and hedging purposes.

2.2 Scope

The model prices single-name PRS contracts with the following characteristics:

- Quarterly premium payments (CDS-standard schedule).
- Single trigger event (flood occurrence at a defined severity level).

- Zero recovery rate (100% loss given default).
- Flat risk-free discount curve.
- Tenors of 1 to 30 years (typically 5 years).

The model does not price:

- Portfolio or index PRS tranches,
- Contracts with variable or stochastic recovery,
- Contracts with counterparty credit risk adjustment (CVA/DVA),
- Contracts with non-standard payment frequencies.

2.3 Trigger Levels

Three trigger levels are supported, corresponding to Environment Agency flood warning severity classifications:

Trigger	Depth Threshold	Description
<code>any_flood</code>	> 0 m	Any flood event above zero depth
<code>moderate</code>	> 0.5 m	Moderate flooding; water ingress to ground floor
<code>severe</code>	> 1.0 m	Severe flooding; significant structural damage

Table 1: PRS trigger level definitions.

3 Mathematical Framework

3.1 Analytical Pricing (`prs_analytical.py`)

The analytical pricing model follows a standard CDS pricing framework adapted for physical flood risk. The key insight is that flood occurrence can be modelled as a Poisson point process, where the survival function takes the same exponential form as in reduced-form credit models.

3.1.1 Hazard Rate Conversion

The upstream GEV hazard curve model provides an annual flood probability $\lambda_{\text{annual}} \in (0, 1)$, representing the probability of at least one flood event exceeding the trigger threshold in any given year. This is converted to a continuous-time Poisson intensity λ via:

$$\lambda = -\ln(1 - \lambda_{\text{annual}}) \quad (1)$$

This transformation ensures consistency between the discrete annual probability and the continuous-time survival function. For small λ_{annual} , the approximation $\lambda \approx \lambda_{\text{annual}}$ holds, but the exact transformation is used throughout.

To prevent numerical overflow, λ_{annual} is capped at 0.999 before conversion.

3.1.2 Survival and Discount Functions

Under the constant hazard rate assumption, the survival probability at time t is:

$$S(t) = \exp(-\lambda t) \quad (2)$$

The risk-free discount factor at time t under continuous compounding is:

$$D(t) = \exp(-r t) \quad (3)$$

where r is the risk-free rate (default $r = 0.03$).

3.1.3 Quarterly Discretisation

The contract is discretised into quarterly periods of length $\Delta t = 0.25$ years. For a contract of tenor T years, the number of periods is $n = 4T$, and the i -th payment date is $t_i = i \cdot \Delta t$ for $i = 1, \dots, n$.

3.1.4 Risky Annuity (Premium Leg)

The risky annuity represents the expected present value of receiving 1 bp per annum, paid quarterly, conditional on no trigger event having occurred:

$$A = \sum_{i=1}^n \Delta t \cdot S(t_i) \cdot D(t_i) \quad (4)$$

This is the denominator in the fair spread calculation and represents the “price” of the premium leg per unit of spread.

3.1.5 Protection Leg

The protection leg represents the expected present value of receiving $(1 - R)$ upon a trigger event, where R is the recovery rate. The probability of a trigger event occurring in the interval $(t_{i-1}, t_i]$ is $S(t_{i-1}) - S(t_i)$, and the payment is assumed to occur at the midpoint of the interval:

$$P = \sum_{i=1}^n (1 - R) \cdot [S(t_{i-1}) - S(t_i)] \cdot D\left(t_i - \frac{\Delta t}{2}\right) \quad (5)$$

where $t_0 = 0$ and $S(t_0) = 1$.

3.1.6 Fair Spread

The fair spread is the premium rate s^* that equates the present value of the premium leg and the protection leg:

$$s^* \cdot A = P \quad \implies \quad s^* = \frac{P}{A} \quad (6)$$

The result is converted to basis points by multiplying by 10,000:

$$s_{\text{bps}}^* = \frac{P}{A} \times 10,000 \quad (7)$$

A floor is applied to ensure the spread does not fall below the minimum PRS spread:

$$s_{\text{final}} = \max(s_{\text{bps}}^*, s_{\text{min}}) \quad (8)$$

where $s_{\text{min}} = 2$ bp (the MIN_PRS_SPREAD_BPS parameter, calibrated to the FloodRE minimum insurance rate).

3.1.7 Fallback Pricing

If the risky annuity $A \leq 0$ (which can occur for extremely high hazard rates where survival probability is negligible), the model falls back to a simple linear approximation:

$$s_{\text{fallback}} = \lambda_{\text{annual}} \times 10,000 \text{ bps} \quad (9)$$

3.2 Recovery Rate

The recovery rate is hardcoded at $R = 0$ for all trigger levels, corresponding to 100% loss given default ($\text{LGD} = 1 - R = 1$). Upon a trigger event the full notional is paid out by the protection seller. No partial recovery is modelled.

3.3 Basis Waterfall

The basis waterfall adjusts gauge-level PRS spreads to property-level spreads. It is defined as a five-component additive model per the PRS Property Pricing Specification (Section 4):

$$B_{\text{total}} = B_{\text{model}} + B_{\text{distance}} + B_{\text{elevation}} + B_{\text{terrain}} + B_{\text{composition}} \quad (10)$$

The property-level PRS spread is then:

$$s_{\text{property}} = s_{\text{gauge}} + B_{\text{total}} \quad (11)$$

where s_{gauge} is the gauge-level fair spread from Equation (7) and B_{total} is the total basis in basis points. A positive basis means the property is riskier than the gauge reference; a negative basis means the property benefits from natural protection relative to the gauge.

3.3.1 Component 1: Model Uncertainty

A fixed additive charge reflecting irreducible model uncertainty in translating gauge-level risk to property-level risk:

$$B_{\text{model}} = +2 \text{ bp (fixed)} \quad (12)$$

3.3.2 Component 2: Distance Basis

The distance basis reflects the reduced spatial correlation between gauge flood readings and property flood exposure as distance increases. It uses inverse-distance weighting (IDW) when multiple gauges are available.

Given N nearest gauges with distances $d_1, d_2, \dots, d_N$ (in kilometres, with a minimum of 0.1 km to avoid singularities), the IDW weights are:

$$w_j = \frac{1/d_j}{\sum_{k=1}^N 1/d_k}, \quad j = 1, \dots, N \quad (13)$$

The weighted distance is:

$$d_w = \min \left(\sum_{j=1}^N w_j \cdot d_j, d_{\text{cap}} \right) \quad (14)$$

where $d_{\text{cap}} = 12$ km. The distance basis is then:

$$B_{\text{distance}} = \min(B_{\text{dist,max}}, \beta_d \cdot d_w) \quad (15)$$

with $\beta_d = 0.5$ bp/km and $B_{\text{dist,max}} = 6$ bp.

3.3.3 Component 3: Elevation Basis

Properties elevated above their reference gauge benefit from reduced flood exposure, reflected as a negative basis (i.e. a reduction in spread). The elevation differential is computed using IDW-weighted gauge elevations:

$$\Delta h = h_{\text{property}} - \sum_{j=1}^N w_j \cdot h_{\text{gauge},j} \quad (16)$$

The elevation basis applies only when the property is above the gauge ($\Delta h > 0$):

$$B_{\text{elevation}} = \begin{cases} \max(-B_{\text{elev,max}}, -\beta_e \cdot \Delta h) & \text{if } \Delta h > 0 \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

with $\beta_e = 0.2$ bp/m and $B_{\text{elev,max}} = 3$ bp. The maximum benefit is therefore -3 bp.

Design note: Properties below the gauge elevation receive no additional penalty via this component. The rationale is that the gauge hazard rate already incorporates the gauge's own elevation context, and additional penalisation of low-lying properties is handled implicitly through the flood zone terrain component.

3.3.4 Component 4: Terrain Variation

The terrain component adjusts for the Environment Agency (EA) flood zone classification of the property location:

EA Flood Zone	Basis (bp)	Description
Zone 3b	+3.0	Functional floodplain
Zone 3a	+3.0	High probability (1 in 100 year or greater)
Zone 3	+3.0	High probability (generic)
Zone 2	+2.0	Medium probability (1 in 100 to 1 in 1,000 year)
Zone 1	0.0	Low probability (<1 in 1,000 year)
Enhanced Defence	-1.0	Behind formal flood defences

Table 2: Terrain basis by EA flood zone classification (Spec Section 4.4).

3.3.5 Component 5: Property Composition

The composition component accounts for property-specific vulnerability characteristics:

Property Type	Basis (bp)
Flat	+2.0
Bungalow	+1.0
Mid-terrace	+0.5
End-terrace	+0.5
Semi-detached	0.0
Detached	0.0

Table 3: Composition basis by property type (Spec Table 3).

An additional +1 bp is applied for properties constructed before the year 2000, reflecting typically lower flood resilience standards in older construction:

$$B_{\text{composition}} = B_{\text{type}} + \begin{cases} 1.0 \text{ bp} & \text{if construction year} < 2000 \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

3.4 QuantLib Pricing (`prs_with_hazard_curve.py`)

An alternative pricing path is provided using the QuantLib library's CDS framework. This serves as an independent cross-validation of the analytical model.

3.4.1 Curve Construction

The QuantLib path supports two modes:

1. **Term structure mode:** A survival probability term structure is loaded from the hazard curve output. An implied constant hazard rate is extracted from the terminal survival probability:

$$\hat{\lambda} = \frac{-\ln S(T)}{T} \quad (19)$$

and used to construct a `FlatHazardRate` curve (required for ISDA CDS engine compatibility).

2. **Flat hazard mode:** A `FlatHazardRate` curve is constructed directly from the annual hazard rate λ_{annual} .

3.4.2 CDS Instrument Construction

The PRS is modelled as a QuantLib `CreditDefaultSwap` with the following conventions:

- **Calendar:** TARGET
- **Day count:** Actual/360 (premium leg)
- **Payment frequency:** Quarterly
- **Date generation:** TwentiethIMM (standard CDS roll dates)
- **Business day convention:** Following
- **Protection start:** $T + 1$ business day
- **Upfront settlement:** $T + 3$ business days
- **Settles accrual:** Yes
- **Pays at default time:** Yes

3.4.3 Pricing Engine

The instrument is priced using the `IsdaCdsEngine` (also known as the “Integral CDS Engine”), which evaluates the fair spread by equating the risk-neutral expectation of premium and protection legs. The risk-free curve is a `FlatForward` curve with annual compounding.

The QuantLib fair spread is obtained via:

```
1 cds.fairSpread() # Returns annualised spread as a decimal
```

and converted to basis points by multiplying by 10,000.

4 Input Parameters and Calibration

4.1 Pricing Parameters

Parameter	Symbol	Default	Description
Annual hazard rate	λ_{annual}	—	From GEV hazard model
Tenor	T	5 years	Contract maturity
Recovery rate	R	0.0	Zero recovery (full LGD)
Risk-free rate	r	0.03	Continuous annual rate
Minimum spread	s_{min}	2.0 bp	FloodRE floor
Quarterly period	Δt	0.25	Fixed quarterly

Table 4: Core pricing parameters.

4.2 Basis Waterfall Parameters

Parameter	Symbol	Value	Description
Model uncertainty	B_{model}	2.0 bp	Fixed charge
Distance rate	β_d	0.5 bp/km	Linear distance coefficient
Distance cap	d_{cap}	12 km	Maximum distance considered
Distance max basis	$B_{\text{dist,max}}$	6.0 bp	Upper bound on distance basis
Elevation rate	β_e	0.2 bp/m	Per-metre benefit rate
Elevation max benefit	$B_{\text{elev,max}}$	3.0 bp	Maximum elevation benefit
Construction year cutoff	—	2000	Pre-2000 adds 1 bp
IDW floor	—	0.1 km	Minimum distance for IDW

Table 5: Basis waterfall parameters.

4.3 Basis Waterfall Inputs

Input	Units	Source
distance_m	metres	Haversine distance from property to gauge
gauge_elevation_m	mAOD	Gauge station elevation (from gauge CDM)
prop_elevation	mAOD	Property ground elevation (from property CDM)
flood_zone	string	EA flood zone classification
property_type	string	Property type (Detached, Flat, etc.)
construction_year	integer	Year of original construction

Table 6: Basis waterfall input data.

4.4 Calibration

The model parameters are calibrated as follows:

- **Hazard rate:** Derived from the GEV hazard curve model (see separate documentation). Not a free parameter of this model.
- **Recovery rate:** Hardcoded at zero (full loss given default). No recovery is assumed upon trigger event.
- **Risk-free rate:** Set to 3% as a representative long-term UK gilt rate. Not dynamically calibrated to market curves.
- **Minimum spread:** Calibrated to the FloodRE Levy 2 minimum insurance premium rate.
- **Basis parameters:** Set by expert judgement and cross-referenced against published EA flood risk data and FloodRE indicative rates.

5 Implementation Details

5.1 Source Files

File	Responsibility
<code>src/models/hazard/prs_analytical.py</code>	Analytical PRS pricing and basis waterfall computation. No external dependencies beyond the Python standard library.
<code>src/models/prs_with_hazard_curve.py</code>	QuantLib-based PRS pricing using the ISDA CDS engine. Requires QuantLib-Python.

Table 7: Source file inventory.

5.2 Two Pricing Paths

The system provides two independent pricing implementations:

Analytical path (`compute_prs_spread`): A pure Python implementation that computes the risky annuity and protection leg via explicit summation over quarterly periods. This is the primary production path, used for all portfolio-level pricing and report generation. It has no dependency on QuantLib.

QuantLib path (`price_prs`): A QuantLib-based implementation that constructs a `CreditDefaultSwap` instrument and prices it using the `IsdaCdsEngine`. This path is used for model validation and produces additional outputs (NPV, premium leg NPV, protection leg NPV, fair upfront).

5.3 Key Implementation Details

5.3.1 Hazard Rate Clamping

The annual hazard rate is clamped to $[0, 0.999]$ before conversion to the continuous rate. A rate of zero returns a spread of 0 bp immediately (no floor applied, as a zero hazard rate means no risk). The upper clamp prevents $\ln(0)$ in Equation (1).

5.3.2 IDW Weighting

Inverse-distance weighting is applied across all nearest gauges to compute a single weighted distance and weighted gauge elevation. The minimum distance floor of 0.1 km prevents singularities when a property is co-located with a gauge. If no gauge data is provided, default values are used (distance = 1.0 km, gauge elevation = property elevation).

5.3.3 Rounding

All basis components are rounded to one decimal place before output. The total basis is the sum of rounded components, not the rounding of the sum, which may introduce up to ± 0.25 bp of rounding error in the total.

5.3.4 QuantLib Date Conventions

The QuantLib implementation uses `ql.Date.todayDate()` as the evaluation date, TARGET calendar, Actual/360 day count for the premium leg, and Actual/365 Fixed for the hazard and discount curves. The schedule is generated with TwentiethIMM date generation, following standard CDS market conventions.

6 Sensitivity Analysis

This section analyses the sensitivity of the PRS fair spread to each key input parameter.

6.1 Hazard Rate Sensitivity

The fair spread is monotonically increasing in the annual hazard rate. For typical UK flood probabilities ($\lambda_{\text{annual}} \in [0.001, 0.10]$), the relationship is approximately linear for low hazard rates and becomes concave for higher rates due to the survival probability declining faster.

Table 8: PRS spread sensitivity to annual hazard rate ($T = 5\text{yr}$, $R = 0$, $r = 0.03$).

λ_{annual}	λ (continuous)	Spread (bp)
0.0010	0.0010	10.0
0.0050	0.0050	50.3
0.0100	0.0100	101.0
0.0200	0.0202	203.3
0.0500	0.0513	518.2
0.1000	0.1054	1,071.6

Table 9: PRS spread sensitivity to risk-free rate ($\lambda_{\text{annual}} = 0.01$, $R = 0$, $T = 5\text{yr}$).

r	Spread (bp)
0.0000	100.6
0.0100	100.8
0.0300	101.0
0.0500	101.3
0.1000	101.9

Table 10: PRS spread sensitivity to tenor ($\lambda_{\text{annual}} = 0.01$, $R = 0$, $r = 0.03$).

Tenor (yr)	Spread (bp)
1	101.0
2	101.0
3	101.0
5	101.0
7	101.0
10	101.0
20	101.0
30	101.0

6.2 Recovery Rate Sensitivity

The fair spread is proportional to $(1 - R)$ in the protection leg numerator. A higher recovery rate reduces the spread linearly. For a fixed hazard rate of $\lambda_{\text{annual}} = 0.01$:

$$\frac{\partial s^*}{\partial R} = -\frac{1}{A} \sum_{i=1}^n [S(t_{i-1}) - S(t_i)] \cdot D\left(t_i - \frac{\Delta t}{2}\right) < 0 \quad (20)$$

6.3 Risk-Free Rate Sensitivity

An increase in the risk-free rate r reduces both the risky annuity and the protection leg PV via higher discount factors. The net effect on the spread ratio P/A is typically small and slightly positive (spread increases with r), as the annuity decreases faster than the protection leg.

For $\lambda_{\text{annual}} = 0.01$, $R = 0$, $T = 5$, see tables above.

6.4 Tenor Sensitivity

Longer tenors produce marginally higher spreads for a given hazard rate, as the conditional default probability per period is constant but the accumulated survival probability decreases. However, under a flat hazard curve, the fair spread converges rapidly and is largely tenor-insensitive for tenors beyond 3–5 years.

6.5 Distance Sensitivity

The distance basis increases linearly at $\beta_d = 0.5$ bp/km up to the cap of 6 bp at 12 km. For a single gauge:

$$B_{\text{distance}}(d) = \min(6.0, 0.5 \times d) \text{ bp}, \quad d \in [0, 12] \text{ km} \quad (21)$$

6.6 Elevation Sensitivity

The elevation basis is a negative (beneficial) adjustment for properties above their reference gauge, increasing in magnitude at 0.2 bp per metre up to a maximum benefit of -3 bp (at 15 m above gauge):

$$B_{\text{elevation}}(\Delta h) = \begin{cases} \max(-3.0, -0.2 \times \Delta h) & \text{if } \Delta h > 0 \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

7 Validation and Backtesting

7.1 Cross-Validation: Analytical vs QuantLib

The primary validation approach is cross-validation between the two independent pricing paths. For flat hazard rate inputs, the analytical and QuantLib implementations are expected to agree to within 0.5 bp for typical parameterisations.

Methodology: For a grid of hazard rates $\lambda_{\text{annual}} \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.10\}$, tenors $T \in \{1, 3, 5, 10\}$ years, and recovery $R = 0$, both pricing paths are evaluated and the absolute spread difference is recorded.

Expected result: Differences arise from:

- Day count convention differences (Actual/365 vs Actual/360 on the premium leg),
- Date generation differences (exact quarterly vs TwentiethIMM roll dates),
- Accrual and settlement date flow handling in the ISDA engine.

These differences are expected to be small (< 1 bp) and do not represent model risk.

7.2 Basis Decomposition Sanity Checks

The basis waterfall is validated through the following checks:

- **Monotonicity:** Distance basis is non-decreasing in distance.
- **Sign correctness:** Elevation basis is non-positive (benefit only).
- **Bounds:** Each component respects its documented cap/floor.
- **Additivity:** Total basis equals the sum of individual components.
- **Range:** For typical Thames catchment properties, total basis falls within $[-1, +14]$ bp.

7.3 Comparison with FloodRE Indicative Rates

As an external benchmark, the model's property-level PRS spreads are compared against FloodRE Levy 2 indicative insurance premium rates for properties in equivalent flood zones. The PRS spread is expected to be of similar order of magnitude to the FloodRE rate, adjusted for differences in:

- Contract structure (swap vs insurance),
- Loss given default assumptions,
- Expense loading (FloodRE includes administrative costs; PRS does not).

Exact numerical agreement is not expected. The comparison serves as a reasonableness check only.

8 Model Limitations and Known Weaknesses

8.1 Constant Hazard Rate Assumption

The analytical model assumes a constant (time-homogeneous) hazard rate over the full contract tenor. This means:

- **No climate trend:** The hazard rate does not increase over time to reflect projected increases in flood frequency due to climate change. For longer tenors (> 5 years), this may understate the true fair spread.
- **No seasonality:** Flood risk is not adjusted for seasonal variation (winter vs summer). The annual hazard rate is an average over the full year.
- **No term structure:** The QuantLib path supports a piecewise survival curve, but the analytical path does not.

8.2 No Counterparty Credit Risk

The model prices the PRS on a risk-free basis with no credit valuation adjustment (CVA) or debit valuation adjustment (DVA). In practice, the protection seller's creditworthiness affects the true economic value of the contract.

8.3 Quarterly Discretisation Error

The continuous-time protection leg is approximated by quarterly discretisation, with default assumed to occur at the midpoint of each quarter. For low hazard rates (typical of UK flood risk), this approximation is accurate. For very high hazard rates ($\lambda_{\text{annual}} > 0.20$), the quarterly discretisation may introduce errors of up to 2–3 bp relative to continuous-time pricing.

8.4 Flat Risk-Free Curve

The risk-free discount curve is flat at the specified rate. In practice, the UK gilt curve has a term structure that affects the present value of future cash flows. For a 5-year PRS, the difference between flat and market-calibrated discounting is typically small (< 1 bp effect on fair spread).

8.5 Additive Basis Model

The five-component basis waterfall is purely additive with no interaction terms. In reality, distance and elevation effects may interact (e.g. a property that is both far from the gauge and at high elevation may have a non-linearly different risk profile). The additive assumption is a first-order approximation.

8.6 No Correlation Modelling

The model prices single-name PRS contracts independently. It does not model spatial correlation of flood events across multiple properties or gauges, which would be required for portfolio-level risk aggregation or tranche pricing.

8.7 Recovery Rate Assumptions

The recovery rate is fixed at $R = 0$ (zero recovery, full loss given default) for all trigger levels and does not vary with property value, flood depth, or construction standards. This is a conservative assumption that maximises the fair spread for a given hazard rate.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

“Does the model do what it says on the tin?”

Yes. The model prices Physical Risk Swap contracts by computing fair annual spreads in basis points using a CDS-equivalent quarterly premium framework. It delivers: (i) gauge-level PRS pricing from hazard curve term structures, (ii) property-level pricing via the five-component basis waterfall (model uncertainty, distance, elevation, terrain, property composition), (iii) basis decomposition for hedging analysis. Both the analytical and QuantLib pricing paths produce consistent results.

9.2 Q2 — Accuracy and Stability

“Does the model provide timely, accurate and stable results?”

The analytical pricing engine produces identical results to the QuantLib ISDA CDS engine to within rounding tolerance for flat hazard rate inputs. The five-component basis waterfall is additive and deterministic. Spreads are monotonically increasing in hazard rate, ensuring economic consistency. Sensitivity analysis confirms stable behaviour across the full range of input parameters. Pricing completes in under 1 ms per contract (analytical path) or under 10 ms (QuantLib path).

9.3 Q3 — Resource Usage

“Does the model consume excessive computational resources?”

No. The analytical pricing path is $O(1)$ per contract with no external dependencies. The QuantLib path requires the QuantLib library but remains lightweight ($O(N_{\text{periods}})$ per contract where N_{periods} is the number of quarterly payment dates). Full portfolio pricing (1,000 contracts) completes in under 1 second on either path. Memory usage is negligible.

9.4 Q4 — Override Controls

“Are there controls over manual overrides?”

The model does not support manual override of computed fair spreads. Contract parameters (tenor, recovery rate, risk-free rate, trigger level) are specified at invocation and are not modifiable post-computation. The MRC can apply a risk rating override via the governance dashboard, but this affects the model’s governance status, not its computational outputs. Basis waterfall components are defined as fixed constants; changes require code modification.

9.5 Q5 — Output Consumption

“How is the output consumed for risk decision-making?”

PRS spreads are the primary output used for: (a) swap pricing and quotation in the Physical Risk market; (b) mortgage credit risk adjustment via flood-adjusted hazard rates; (c) basis risk analysis for hedging effectiveness assessment; (d) stress testing via scenario-dependent hazard curves. Results are displayed in the PRS pricing panel of the interactive governance dashboard and exported in PDF property reports.

9.6 Q6 — Model Chaining

“Is the model’s output used as input to another model?”

Yes. Downstream, PRS spreads feed the Mortgage Credit Risk Pricer (MKM-MORT-001) for flood-adjusted expected loss calculation. Basis analysis outputs inform hedging strategy decisions. Upstream, the model consumes term structures from the GEV Hazard Curve Model (MKM-HAZARD-001) and property-level hazard rates from the Property Hazard Curve module. The model also depends indirectly on the Property Valuation Engine (MKM-PROP-001) for basis waterfall inputs.

9.7 Q7 — Weaknesses

“Does the model have weaknesses and limitations of use?”

Yes; the principal weaknesses are documented in Section 8. Key limitations: (i) flat hazard rate assumption over the contract tenor (no time-varying climate risk); (ii) additive basis waterfall with no interaction terms between components; (iii) quarterly premium schedule is CDS-standard and may not match bespoke contract terms; (iv) the recovery rate is treated as a fixed constant rather than a stochastic variable. The model prices single-name contracts only; portfolio-level PRS structuring is out of scope.

9.8 Q8 — Monitoring

“What type of monitoring is required?”

The following monitoring is recommended: (a) quarterly comparison of model spreads against any available market transaction data; (b) annual review of basis waterfall component calibration against property flood loss experience; (c) cross-validation between analytical and QuantLib pricing paths following library updates; (d) sensitivity analysis of spread to recovery rate and risk-free rate assumptions; (e) stress testing of spreads under all five intensity scenario families.

9.9 Q9 — Overall Risk

“Is the overall model risk acceptable?”

Conditionally acceptable. As a Tier 1 model with direct pricing impact, the PRS Pricing Engine carries significant model risk. The flat hazard rate assumption is the primary concern: under climate change scenarios, hazard rates are likely to increase over multi-year tenors, leading to underpriced protection. This is partially mitigated by conservative scenario selection and the basis waterfall safety margin. The interaction-free basis waterfall is a known simplification accepted by the MRC pending future enhancement.

10 Change History

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions (Chapter 5)	D. Kelly